HDOC - Day 17 Activity

C Programming Assignment

Name-Arushi Gusain

Sap id-590033489

Batch 15

Date- 10/09/2026

Menu-Driven Digit Operations using switch-case and User-Defined Functions

Problem Requirements

Write a menu-driven C program for a positive integer. Use switch-case. Each menu option must call a separate void user-defined function with a simple function call. Each function reads its own input, processes it, and displays the result.

1. Generate the largest possible number using its digits.
2. Generate the smallest possible number using its digits.
3. Swap its first and last digits.
4. Check whether it is a palindrome number.

Algorithm

Main Algorithm

1. Start and display the four menu options.
2. Read choice.
3. Use switch(choice).
4. Choice 1: call largestNumber().
5. Choice 2: call smallestNumber().
6. Choice 3: call swapFirstLast().
7. Choice 4: call palindromeCheck().
8. Otherwise display Invalid choice.
9. Stop.

Function largestNumber()

1. Read a positive integer n and validate it.
2. Initialize freq[10] to zero.
3. Extract digits using n % 10 and count each digit.
4. Print digits from 9 down to 0 according to their frequencies.
5. The printed sequence is the largest possible number.

Function smallestNumber()

1. Read and validate n; count the frequency of digits 0 to 9.
2. Find and print the smallest non-zero digit first.
3. Print all zeroes after the first digit.
4. Print remaining digits from 1 to 9 in ascending order.
5. This avoids an invalid leading zero and gives the smallest number.

Function swapFirstLast()

1. Read and validate n.
2. For a single-digit number, display the same number.
3. Find the highest power of 10 in n.
4. Find first digit, last digit and the middle part.
5. Form the new value after interchanging the first and last positions.
6. Display the swapped integer.

Function palindromeCheck()

1. Read and validate n.
2. Store n in temp and set reverse = 0.
3. Reverse the number digit by digit.
4. Compare reverse with the original number.
5. If equal print Palindrome; otherwise print Not Palindrome.

Test Case Table

Case	Choice	Input	Expected Output	Status
1	1	31052	Largest possible number = 53210	Pass
2	4	12321	12321 is a palindrome number	Pass
3	2	31052	Smallest possible number = 10235	Pass
4	2	10020	Smallest possible number = 10002	Pass
5	3	12345	After swapping = 52341	Pass
6	5	-	Invalid choice	Pass

C Program

```
#include <stdio.h>
void largestNumber(void);
void smallestNumber(void);
void swapFirstLast(void);
void palindromeCheck(void);
int main(void)
{
int choice;
printf("DIGIT OPERATIONS MENU\n");
printf("1. Generate largest possible number\n");
printf("2. Generate smallest possible number\n");
printf("3. Swap first and last digits\n");
printf("4. Check whether the number is palindrome\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice)
{
case 1: largestNumber(); break;
case 2: smallestNumber(); break;
case 3: swapFirstLast(); break;
case 4: palindromeCheck(); break;
default: printf("Invalid choice.\n");
}
return 0;
}
void largestNumber(void)
{
long long n, temp;
int freq[10] = {0}, digit, i;
printf("Enter a positive integer: ");
scanf("%lld", &n);
if (n <= 0) {
printf("Invalid input. Enter a positive integer.\n");
return;
```

```
}
temp = n;
while (temp > 0) {
digit = temp % 10;
freq[digit]++;
temp /= 10;
}
printf("Largest possible number = ");
for (i = 9; i >= 0; i--)
while (freq[i] > 0) {
printf("%d", i);
freq[i]--;
}
printf("\n");
}
void smallestNumber(void)
{
long long n, temp;
int freq[10] = {0}, digit, i, first = -1;
printf("Enter a positive integer: ");
scanf("%lld", &n);
if (n <= 0) {
printf("Invalid input. Enter a positive integer.\n");
return;
}
temp = n;
while (temp > 0) {
digit = temp % 10;
freq[digit]++;
temp /= 10;
}
for (i = 1; i <= 9; i++)
if (freq[i] > 0) {
first = i;
break;
```

```
}
printf("Smallest possible number = ");
printf("%d", first);
freq[first]--;
while (freq[0] > 0) {
printf("0");
freq[0]--;
}
for (i = 1; i <= 9; i++)
while (freq[i] > 0) {
printf("%d", i);
freq[i]--;
}
printf("\n");
}
void swapFirstLast(void)
{
long long n, power = 1;
long long first, last, middle, result;
printf("Enter a positive integer: ");
scanf("%lld", &n);
if (n <= 0) {
printf("Invalid input. Enter a positive integer.\n");
return;
}
if (n < 10) {
printf("After swapping = %lld\n", n);
return;
}
while (n / power >= 10)
power *= 10;
first = n / power;
last = n % 10;
middle = (n % power) / 10;
result = last * power + middle * 10 + first;
```

```
printf("After swapping = %lld\n", result);
}
void palindromeCheck(void)
{
long long n, temp, reverse = 0;
int digit;
printf("Enter a positive integer: ");
scanf("%lld", &n);
if (n <= 0) {
printf("Invalid input. Enter a positive integer.\n");
return;
}
temp = n;
while (temp > 0) {
digit = temp % 10;
reverse = reverse * 10 + digit;
temp /= 10;
}
if (reverse == n)
printf("%lld is a palindrome number.\n", n);
else
printf("%lld is not a palindrome number.\n", n);
}
```

Compilation , Execution & Testing Cases

```
Session  Actions  Edit  View  Help                                kali@kali: ~
┌──(kali㉿kali)-[~]
└─$ vi num1.c

┌──(kali㉿kali)-[~]
└─$ cc num1.c

┌──(kali㉿kali)-[~]
└─$ ./a.out
DIGIT OPERATIONS MENU
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check whether the number is palindrome
Enter your choice: 1
Enter a positive integer: 31052
Largest possible number = 53210

┌──(kali㉿kali)-[~]
└─$ ./a.out
DIGIT OPERATIONS MENU
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check whether the number is palindrome
Enter your choice: 4
Enter a positive integer: 12321
12321 is a palindrome number.

┌──(kali㉿kali)-[~]
└─$ ./a.out
DIGIT OPERATIONS MENU
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check whether the number is palindrome
Enter your choice: 2
Enter a positive integer: 31052
Smallest possible number = 10235
```

```
┌──(kali㉿kali)-[~]
└─$ ./a.out
DIGIT OPERATIONS MENU
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check whether the number is palindrome
Enter your choice: 2
Enter a positive integer: 10020
Smallest possible number = 10002

┌──(kali㉿kali)-[~]
└─$ ./a.out
DIGIT OPERATIONS MENU
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check whether the number is palindrome
Enter your choice: 3
Enter a positive integer: 12345
After swapping = 52341

┌──(kali㉿kali)-[~]
└─$ ./a.out
DIGIT OPERATIONS MENU
1. Generate largest possible number
2. Generate smallest possible number
3. Swap first and last digits
4. Check whether the number is palindrome
Enter your choice: 5
Invalid choice.

┌──(kali㉿kali)-[~]
└─$ █
```